



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Undergraduate Internship Report

Program: B.Tech
Department: AI & DS
Academic Year: 2025-2026

Internship Duration: 15-11-2025 TO 15-1-2026
Name of Student: Manjunadha.D
Register Number: 192324237
Batch: 2023-2027
Internship Organization: Inspire Ai
Supervisor in Company: Pravallika

1. Internship Completion Certificate



CERTIFICATE OF INTERNSHIP COMPLETION



This Certificate is Proudly Presented To

Manjunadha.D

In appreciation of your valuable contribution and continuous professional growth in the specialized domain of Artificial Intelligence during your internship with INSPIRE AI, we are pleased to certify your successful completion of all assigned tasks and projects.

PAVAN KUMAR
FOUNDER & DIRECTOR



CERTIFICATE ID : IAI/ICC1502 |
DURATION: 15/11/2025 to 15/1/2026

2. Student Declaration

I, Manjunadha.D, a student of Artificial intelligence of AI & DS, solemnly affirm that the internship experience documented in this report reflects my direct involvement in a formal organizational setting, actively engaged in industry-relevant operations.

During the internship period from 15-11-2025 to 15-1-2026, I was entrusted with tasks and responsibilities that were integral to the company's ongoing functions. The insights and observations included herein are a true representation of the work culture, technical exposure, and professional interactions I encountered.

This report has been prepared based on actual experiences and contributions during my time with the organization. I have abided by the professional conduct expected of an intern and ensured that the contents of this document are factual, original, and aligned with the intended academic and industrial learning objectives.

Signature of the Student

Date: _____

3. Company Overview

- Name of the Company: Inspire Ai
- Nature of Business/Industry Sector:
Inspire Ai is an educational platform that provides learning modules and various projects for upskilling and career growth in collaboration with partnered companies. Wipro is a Credential Platform Partner..
- Brief History & Overview: :
Inspire Ai is an Indian e-learning and IT consulting **company** that provides online courses, mentorship, and project opportunities in trending technologies to help students and professionals build careers.
- Location(s):
Inspire AI Technology Solutions Private Limited Is V300, Indu Fortune Field, KPHB Phase13, Kukatpally, Madhapur, Hyderabad, Shaikpet, Telangana, India, 500081.
- Products/Services Offered:
Inspire AI offers a diverse portfolio of educational services and programs, which are supported by its flexible learning platform:
Project Access: Offering various real-time or simulation projects to gain practical exposure and potential to grow and learn.
Upskilling & Career Growth Support: Focusing on being a part of the user's upskilling journey and career growth.
Credential Platform Partnership: Operating in collaboration with credential platform partners, such as Wipro.
Orientation Sessions: Conducting preparatory orientation sessions prior to the commencement of the batch for a complete walkthrough of the journey.
- Major Clients/Projects:

1. AI Resume Screening System

Project Goal:

To develop an AI-based system that analyzes resumes and shortlists candidates based on required skills, experience, and job description matching.

Role & Contribution:

Designed a simple web interface to upload resume details. Implemented text processing logic to extract important keywords such as skills, education, and experience. Compared resume content with job requirements and displayed a matching score.

Technologies Used:

Python, HTML, CSS, JavaScript, Natural Language Processing

Outcome:

Successfully created a resume screening model that helps reduce manual effort in candidate selection and improves the speed of recruitment analysis

2. AI Chatbot for Customer Support

Project Goal:

To create an AI chatbot that answers basic customer queries and provides instant support through a web interface.

Role & Contribution:

Developed the chatbot UI using HTML and CSS. Implemented JavaScript logic for user interaction and connected predefined response data to provide quick answers. Added common query handling for services, contact details, and support information.

Technologies Used:

HTML, CSS, JavaScript, Python, NLP

Outcome:

Built an interactive chatbot that provides quick responses to user queries and improves customer support efficiency.

3. Sentiment Analysis on User Reviews

Project Goal:

To build an AI model that analyzes user reviews and classifies them as positive, negative, or neutral.

Role & Contribution:

Collected sample review data and performed text preprocessing such as removing stop words and special characters. Trained a basic sentiment classification model and displayed the result through a simple interface.

Technologies Used:

Python, Machine Learning, NLP, Pandas, Scikit-learn

Outcome:

Successfully developed a sentiment analysis system that helps understand customer opinions and feedback trends.

4. Fake News Detection System

Project Goal:

To develop a machine learning model that identifies whether a news article is real or fake based on its text content.

Role & Contribution:

Prepared a dataset of news articles and applied text preprocessing techniques. Converted text into numerical features using TF-IDF and trained a classification model to predict fake or real news.

Technologies Used:

Python, NLP, Machine Learning, TF-IDF, Scikit-learn

Outcome:

Created a working fake news detection system that helps identify misleading news content and supports safer information sharing.

4. Internship Role and Objectives

- Internship Title/Designation:

As an AI Intern with in Inspire AI their self-paced program, my primary role was to learn, develop, and implement basic Artificial Intelligence and Machine Learning models for real-time problem-solving applications. I worked on data collection, data preprocessing, model training, testing, evaluation, and integrating AI-based features into simple applications.

The internship helped me understand how AI systems are designed and how machine learning models are used to analyze data, identify patterns, and generate predictions. I also gained practical knowledge in Natural Language Processing, classification models, and AI-based application development.

Skills Acquired:

Technical Skills:

- Programming Language:
Python
- AI and Machine Learning Concepts:
Supervised Learning, Classification, Prediction Models
- Data Processing:
Data Cleaning, Data Preprocessing, Feature Extraction
- Machine Learning Libraries:
NumPy, Pandas, Scikit-learn
- Natural Language Processing:
Text Preprocessing, Tokenization, Stop Word Removal, TF-IDF Vectorization
- Model Development:
Model Training, Testing, Accuracy Evaluation
- Data Visualization:
Matplotlib, Seaborn
- AI Application Development:
Building simple AI-based systems for text classification, prediction, and recommendation
- Tools Used:
Google Colab, Jupyter Notebook, VS Code
- Version Control:
Git and GitHub

Organizational & Managerial Skills:

- Collaborated in planning AI project workflows, including dataset selection, preprocessing steps, model training, and testing phases.
- Assisted in preparing project documentation, model workflow diagrams, dataset descriptions, and result analysis.
- Contributed to dividing AI projects into smaller modules such as data collection, cleaning, model building, evaluation, and report preparation.
- Improved the ability to test AI models using sample inputs and evaluate performance using accuracy scores and prediction outputs.

Soft Skills:

- **Analytical Thinking & Problem-Solving:**

Strengthened the ability to analyze real-world problems and convert them into AI-based solutions. Improved skills in understanding datasets, identifying important features, cleaning data, and selecting suitable machine learning models for prediction or classification tasks.

- **Logical Reasoning & Systems Design:**

Enhanced understanding of designing AI workflows, including data collection, preprocessing, feature extraction, model training, testing, and evaluation. Learned how to choose proper algorithms based on the project requirement and expected output.

- **Attention to Detail & Code Quality:**

Gained precision in handling datasets, removing errors, checking missing values, tuning model inputs, and validating prediction results. Improved the ability to write clean Python code with proper structure, comments, and error handling.

- **Effective Documentation & Communication:**

Improved the ability to document AI project workflows clearly, including dataset details, preprocessing steps, model training process, testing results, and prediction outputs. Used code comments, project reports, and GitHub repositories to explain the working of AI models in a simple and understandable way.

- **Project & Time Management:**

successfully managed timelines for multiple self-paced AI projects, from dataset collection and preprocessing to model development, testing, and final documentation. Improved prioritization skills by dividing each project into smaller tasks such as data cleaning, algorithm selection, model training, evaluation, and report preparation..

Objectives of the Internship:

- To gain practical experience in developing AI and Machine Learning based applications, including data collection, data preprocessing, model training, testing, and evaluation.
- To achieve proficiency in essential AI technologies and tools such as Python, NumPy, Pandas, Scikit-learn, Matplotlib, Jupyter Notebook, Google Colab, and VS Code.
- To develop strong problem-solving and analytical abilities required to convert real-world problems into AI-based solutions.
- To understand the complete AI project development lifecycle, including dataset selection, data cleaning, feature extraction, model selection, training, testing, evaluation, and deployment.
- To gain knowledge of Natural Language Processing techniques such as text preprocessing, tokenization, stop word removal, TF-IDF vectorization, and text classification.
- To learn how to evaluate AI models using accuracy score, confusion matrix, precision, recall, and prediction outputs.
- To follow professional development practices, including writing clean and documented Python code, maintaining project files properly, using Git/GitHub for version control, and following ethical AI practices.

5. Work Description

As an AI Intern, I was responsible for developing and implementing basic Artificial Intelligence and Machine Learning models for real-world problem-solving applications. My role involved collecting and preprocessing datasets, selecting suitable algorithms, training models, testing predictions, and evaluating model performance.

I also worked on AI-based projects such as text classification, sentiment analysis, fake news detection, and chatbot development. Through this internship, I gained practical experience in Python programming, data handling, machine learning workflows, model evaluation, and AI project documentation.

Tasks & Key Responsibilities

- **Data Collection:** Collected suitable datasets from reliable sources for AI and Machine Learning based projects.
- **Data Preprocessing:** Cleaned datasets by removing missing values, duplicate records, unwanted symbols, and unnecessary columns.
- **Feature Extraction:** Converted raw data into useful features for model training, especially in text-based projects using NLP techniques.
- **Model Development:** Selected suitable Machine Learning algorithms for classification and prediction-based tasks.
- **Model Training:** Trained AI models using processed datasets and improved performance through repeated testing.
- **Model Evaluation:** Evaluated model performance using accuracy score, confusion matrix, precision, recall, and prediction results.
- **Natural Language Processing:** Applied NLP techniques such as tokenization, stop word removal, text cleaning, and TF-IDF vectorization.

- **Python Programming:** Used Python libraries such as NumPy, Pandas, Scikit-learn, and Matplotlib for data handling, model building, and result visualization.
- **Testing & Debugging:** Tested AI models with sample inputs, identified errors, and improved prediction accuracy.
- **Documentation:** Prepared project documentation, workflow explanations, result analysis, and final internship report content.

Real-Time Projects Participated In:

Project: SmartBot AI — Intelligent Student Assistant

Goal:

To build an AI-powered student assistant that helps students by answering academic questions, providing study guidance, managing common queries, and giving quick learning support.

Implementation:

Developed a simple and user-friendly chatbot interface for students to enter their questions.

Created AI-based response logic to understand student queries related to subjects, assignments, exams, and study materials.

Implemented Natural Language Processing techniques to process user input and identify important keywords.

Designed a response system to provide relevant answers, study tips, and basic academic support.

Tested the chatbot with different student queries to improve response accuracy and user experience.

Prepared project documentation explaining the chatbot workflow, input processing, response generation, and result analysis.

Outcome:

Successfully developed an intelligent student assistant chatbot that provides quick academic support and improves student interaction with learning resources. The project demonstrated practical knowledge of AI chatbot development, NLP concepts, Python programming, user query handling, and AI-based response generation.

Tools & Technologies Use

1. Programming Language: Python
2. AI / Machine Learning: Machine Learning Algorithms, Classification Models, Prediction Models
3. Natural Language Processing: Text Preprocessing, Tokenization, Stop Word Removal, Keyword Extraction
4. Libraries: NumPy, Pandas, Scikit-learn, Matplotlib
5. Development Tools: VS Code, Jupyter Notebook, Google Colab
6. Dataset Handling: CSV Files, Dataset Cleaning, Feature Extraction
7. Model Evaluation: Accuracy Score, Confusion Matrix, Precision, Recall
8. Version Control: Git, GitHub
9. Documentation Tools: MS Word, PowerPoint, GitHub README files

Meetings & Professional Interactions

- **Technical Sprints:** Participated in weekly check-ins to discuss feature roadmap and resolve integration bottlenecks.
- **UI/UX Reviews:** Collaborated in design sessions to refine layouts for better user accessibility and responsiveness.
- **API Documentation:** Documented endpoint structures and request/response cycles to ensure clean collaboration during the development lifecycle.
- **Code Pushes:** Maintained an active development history with daily commits, focusing on readable code and detailed pull request descriptions.

6. Learning Outcomes

Industry Exposure Gained:

- **Professional Standards:**
Gained practical exposure to the AI project development lifecycle, including dataset collection, data preprocessing, model training, testing, evaluation, and documentation.
- **Model Development Workflow:**
Understood how AI models are developed step by step, from selecting a dataset to applying algorithms and checking prediction accuracy.
- **Self-Paced Discipline:**
Developed time-management skills by completing AI-based projects in a self-paced learning environment and meeting project milestones independently.
Skills Acquired:
- **Technical Skills:**
Gained proficiency in Python programming, Machine Learning concepts, Natural Language Processing, data preprocessing, feature extraction, model training, and model evaluation.
- **Tools and Libraries:**

Worked with NumPy, Pandas, Scikit-learn, Matplotlib, Jupyter Notebook, Google Colab, VS Code, Git, and GitHub.

- **Managerial Skills:**
Improved documentation skills by preparing project reports, model workflow explanations, result analysis, and GitHub README files.
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- **Soft Skills:**
Enhanced problem-solving ability, logical thinking, attention to detail, and continuous learning by identifying errors, improving datasets, and testing different model outputs. Understanding of Industrial Workflows:
- **AI Development Workflow:**
Understood the importance of building AI projects in stages, including problem identification, dataset preparation, model selection, training, testing, and improvement.
- **Data-to-Prediction Flow:**
- Gained a clear understanding of how raw data is cleaned, converted into features, passed into a machine learning model, and used to generate predictions.
- **Testing and Evaluation:**
- Learned how model performance is measured using accuracy score, confusion matrix, precision, recall, and prediction results.
- **Contribution to Team/Project:**
 - **End-to-End Implementation:** Successfully developed and deployed AI, reducing manual data entry for mock clients through automation.
 - **Performance Optimization:** Improved frontend loading times by optimizing CSS layouts and streamlining JavaScript logic.
 - **Measurable Impact:** Achieved **100% data persistence** and secure user authentication across all personal projects, ensuring zero data loss during testing phases.
- **Comparison with Academic Learning:**
 - **Theory vs. Practice:** While academics focused on the syntax of HTML/CSS, the internship taught me how to use these tools to solve **real-world business problems** (e.g., Inventory and Food Ordering systems).
 - **Scalability:** Academics often involve small scripts; this internship required thinking about **Scalable Data**, such as how to structure MongoDB to handle hundreds of items or users efficiently.
 - **Debugging:** Shifted from "solving for a grade" to "solving for a user," prioritizing

- bug-free interfaces and smooth navigation.

7. Challenges Faced and Solutions

• Technical Challenges & Resolutions

1. The "Stale Data" Bug

- **Challenge:** The UI wouldn't refresh after adding or deleting items in MongoDB.
- **Resolution:** Implemented **Async/Await** and state-refresh logic to ensure the frontend updates instantly after a successful API response.

2. Layout Breaks on Mobile

- **Challenge:** Forms and navigation menus became unusable on small screens.
- **Resolution:** Switched to a **Mobile-First CSS approach** using Flexbox and Media Queries to ensure 100% responsiveness.

3. API Loading Delays

- **Challenge:** The app would crash while waiting for data from the server.
- **Resolution:** Added **Conditional Rendering** (checking if data exists) and "Loading" states to improve user experience.

• How they were resolved:

- **Implemented Fetch & Callbacks:** Resolved data sync issues by using **Asynchronous JavaScript (Async/Await)** and `fetch()` to ensure the UI only updated after receiving a successful 200 OK status from the server.

Mobile-First Refactoring: Fixed layout breaks by refactoring CSS using **Media Queries** and switching from fixed pixel widths to relative units like REM, VW, and VH, ensuring a fluid grid.

• Feedback from Supervisor

- **Self-Reliance:** "Proactive in troubleshooting backend errors without constant guidance."
- **Architecture:** "Strong understanding of the Request-Response cycle and MongoDB schema design."
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8. Reflection and Conclusion

• Overall Experience

- **Professional Growth:** The internship was a transformative bridge between academic theory and industry reality. Moving from small class assignments to building a **full-stack ecosystem** (Frontend, Backend, and Database) significantly boosted my confidence as a developer.
- **Problem-Solving Mindset:** Beyond writing code, I learned that software development is about solving **user problems**. Debugging complex MongoDB connections and CSS layout breaks taught me patience and analytical thinking.

• Career Relevance

- **Full-Stack Competency:** This experience solidified my expertise in the **MERN stack logic** (minus the React framework), making me a versatile candidate for Junior Developer roles.
- **Direct Alignment:** As an **AI & DS student**, understanding how data is collected and stored via web applications is crucial. This internship provided the "data collection" foundation that complements my "data analysis" studies perfectly.

- **Tool Mastery:** Gained mastery over professional tools like **Git/GitHub, Postman, and MongoDB Atlas**, which are essential for any modern engineering role.

- **Suggestions for Future Interns**

1. **Master the Fundamentals:** Don't rush into frameworks. Spend time understanding **Vanilla JavaScript** and **CSS Flexbox**; they make learning everything else much easier.
2. **Focus on Documentation:** Write your **README** files as you go. It helps you remember your logic and makes your GitHub look professional to recruiters.
3. **Build Projects that Solve Problems:** Instead of just copying tutorials, try to build a project that fixes a small daily issue, like an expense tracker or a personal link-vault.

9. Appendices

The screenshot displays the Visual Studio Code (VS Code) interface. The Explorer panel on the left shows a project structure with a 'public' folder containing 'index.html', 'css', 'js', and 'routes'. The main editor area shows the content of 'index.html', which is an HTML document for 'SmartBot AI - Intelligent Student Assistant'. The document includes a DOCTYPE declaration, HTML lang attribute, charset, viewport, description, keywords, author, title, and links to fonts and a stylesheet. It also contains script tags for 'app.js', 'chat.js', and 'admin.js'. The bottom panel shows the 'TERMINAL' tab with the command 'node server.js' executed. The output indicates 'Database initialized' and 'SmartBot AI Server' running at 'http://localhost:5000' with default admin credentials 'admin / admin123'.

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <meta name="description" content="SmartBot AI - Intelligent conversational assistant for students. Get study h
7   <meta name="keywords" content="AI chatbot, student assistant, study help, NLP, smart bot">
8   <meta name="author" content="SmartBot AI Team">
9   <title>SmartBot AI - Intelligent Student Assistant</title>
10  <link rel="preconnect" href="https://fonts.googleapis.com">
11  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
12  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800;900&family=JetBrains+M
13  <link rel="stylesheet" href="/css/styles.css">
14 </head>
15 <body>
16   <div id="app"></div>
17
18   <script src="/js/app.js"></script>
19   <script src="/js/chat.js"></script>
20   <script src="/js/admin.js"></script>
21 </body>
```

```
> node server.js
[✓] Database initialized
SmartBot AI Server
Running at http://localhost:5000
Default admin: admin / admin123
```

10. Faculty Mentor's Remarks

This is to certify that the internship report submitted by Manjunadha.D has been reviewed and it is evident that the internship was undertaken at a bona fide industry or company. The work documented is satisfactory and relevant to the student's field of study.

Faculty Name & Designation

Signature:

Date: